42.510 Statistical Learning for Data Science

Statistical decision theory

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Week 3, Lecture 1

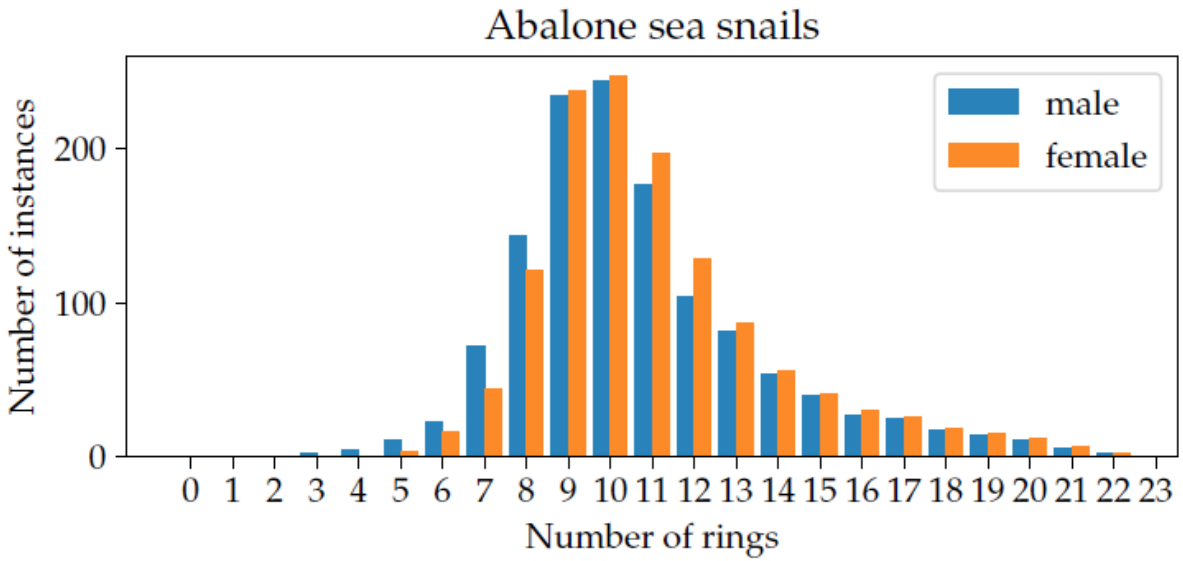
Supervised learning

- Observe n instances of variables $\mathbf{X}, Y$. Predict Y knowing only $\mathbf{X}$.
- Idea: **Prediction** with minimum error.
- **Example:** Predict the gender of **Abalone** from number of rings on their shell.



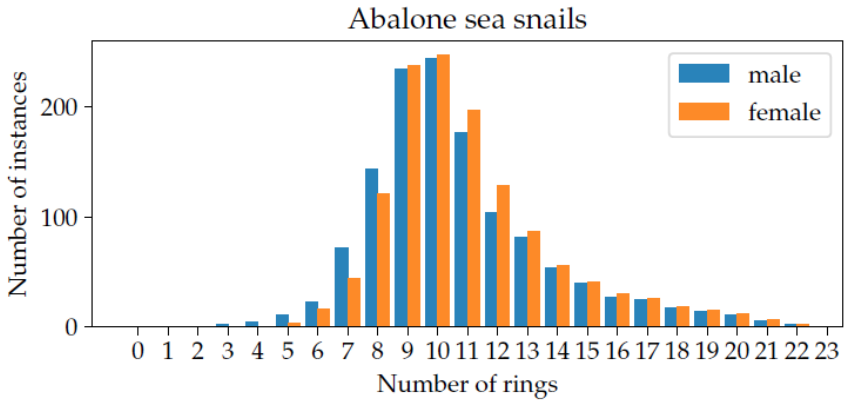
Supervised learning

- **Example:** Predict the gender of Abalone from number of rings on their shell.



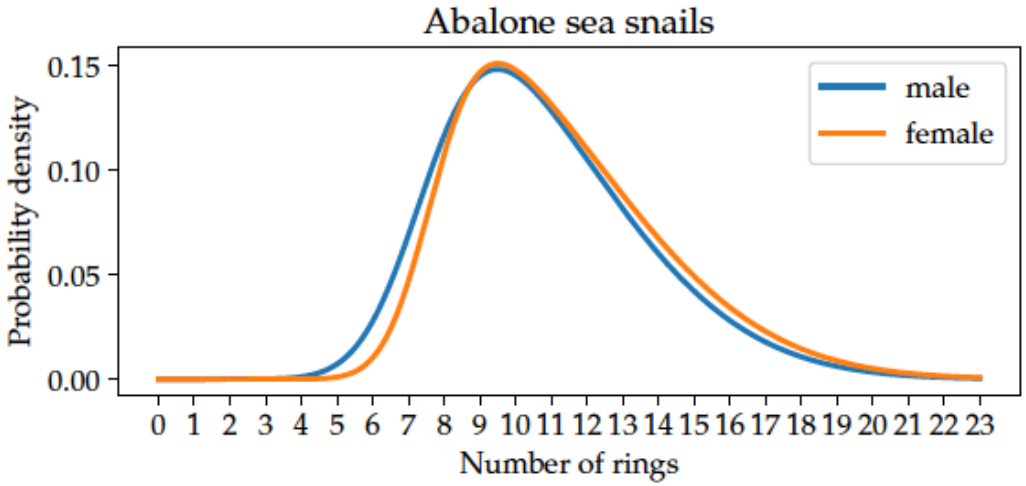
- *Predict* with minimum mistakes:
  - Predict **male** if # rings ≤ 8,
  - Predict **female** if # rings ≥ 9.

Supervised learning



- Observations on the prediction:
  - Still many mistakes (though better than random).
  - We used a **minimum error rule**.
  - Can be applied with more variables, say we include: length of shell.
  - **Issue:** exact computation may need examining entire population.
- Statistics may help:
  - Assume we know the distribution $\mathbb{P}(X, Y)$.
  - In the above example, suppose we know the probability distributions of the number of rings (say, a 3-parameter skew normal) for both genders and the relative proportion of males and females; 7 parameters in total.

Supervised learning

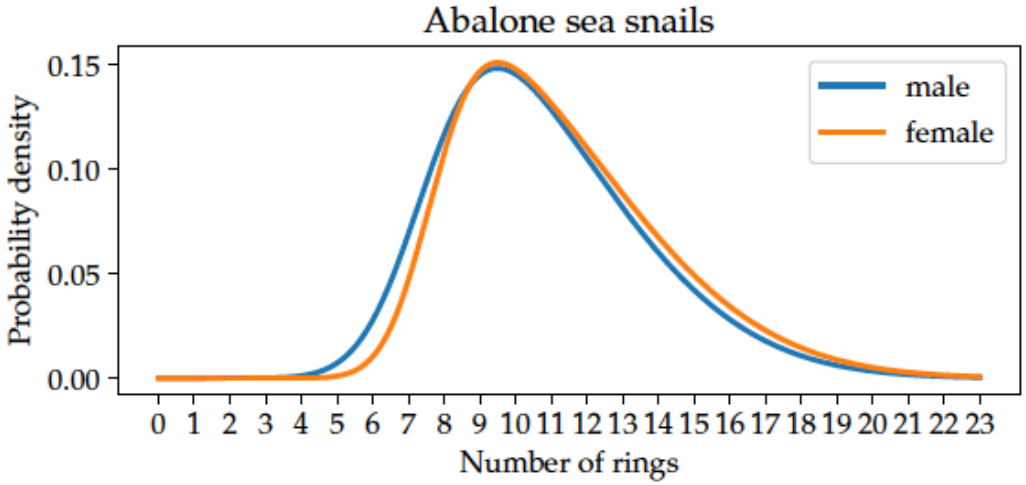


- Denote $Y = 0$ as **male** and $Y = 1$ as **female**.
- Let $p_0 = \mathbb{P}(Y = 0)$ and $p_1 = \mathbb{P}(Y = 1)$.
- Let $\mathbb{P}(x|Y = 0) \sim f_0$ and $\mathbb{P}(x|Y = 1) \sim f_1$.
- Then a natural decision rule is

$$\widehat{Y}(x) = \begin{cases} 1 & \text{if } p_1 f_1(x) > p_0 f_0(x), \\ 0 & \text{otherwise.} \end{cases}$$

- Leads to the same decision rule: Male if # rings ≤ 8, Female otherwise. (assuming $p_0 \approx p_1$ here)
- We used 7 parameters to come up with the prediction rule.

Supervised learning



- Here we assumed knowledge of the population and a particular rule for error/ violation.
- In reality: we do not know $\mathbb{P}(\mathbf{X}, Y)$ and we choose *loss rule* according to the problem.
- Let's see the general set up.

Supervised learning: regression

- **Input: X**, **Output:** Y.
- Output can be quantitative, qualitative (or qualitative but ordinal, which we do not address now).
- Consider **quantitative output** (regression problem).
- Input vector is given by $\mathbf{X} \in \mathbb{R}^p$.
- Output is $Y \in \mathbb{R}$.
- Joint distribution: $\mathbb{P}(\mathbf{X}, Y)$ (often unknown).
- **Goal**: Find $f(\mathbf{X})$ which can predict Y.
- How?
- Find f that minimizes some loss: $L(Y, f(\mathbf{X}))$, the loss/error in predicting Y with $f(\mathbf{X})$.

Loss minimization: squared error loss

- Minimizing **Loss** in an expected sense.
- **Risk** or **Expected Prediction Error** (EPE).

$$R(f) = \mathbb{E}L(Y, f(X)).$$

- One popular loss function is the **Squared Error Loss** where **Risk** or **EPE** is

$$R(f) = \mathbb{E}(Y - f(X))^2 = \int (y - f(x))^2 \mathrm{d}\mathbb{P}(\mathbf{x}, y)).$$

- Using conditioning we get

$$R(f) = \mathbb{E}_{\mathbf{X}} \mathbb{E}_{Y|\mathbf{X}} \left[(Y - f(\mathbf{X}))^2 | \mathbf{X} \right].$$

- Hence we can get a solution at each fixed x:

$$f^*(\mathbf{x}) = \arg\min_c \mathbb{E}_{Y|\mathbf{X}} \left((Y - c)^2 | \mathbf{X} = \mathbf{x} \right).$$

- **Solution:** *regression function*

$$f^*(\mathbf{x}) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}].$$

- This is called the *Bayes predictor* or *Bayes estimator* (in Statistics). And the associated **Risk**, called *Bayes risk*.

Approach 1: Nearest neighbor algorithm

- Data:
$$(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n).$$
- This requires no structural algorithm and approximates the regression solution
$$\widehat{f}(\mathbf{x}) = \text{Ave}(y_i | \mathbf{x}_i \in \mathbf{N}_k(\mathbf{x}))$$
where $\mathbf{N}_k(\mathbf{x})$ = neighborhood of $\mathbf{x}$ with k points.
- Let n = number of training data points.
- We can show that as $n, k \to \infty$ with $k/n \to 0$,
$$\widehat{f}(\mathbf{x}) \to \mathbb{E}[Y | \mathbf{X} = \mathbf{x}].$$
- Unfortunately when dimension $p \to \infty$ this turns out to be disastrous.
  - convergence does happen but very slowly.
- *Data-based approach* (no model/structural assumption) .
- We used local approximation method for the conditional expectation.

Approach 2: Linear regression

- Linear regression: strong structural assumption.

$$f(\mathbf{x}) = \mathbf{x}^\top \beta$$

- Plugging in to the Risk function $R(f)$ and minimizing we obtain

$$\widehat{\beta} = (\mathbb{E}(\mathbf{X}\mathbf{X}^\top))^{-1}\mathbb{E}(\mathbf{X}Y)$$

- Rather than conditioning on $\mathbf{x}$, we pool knowledge over all values.
- Hence we have a global linear function for conditional expectation.
- This is a *model-based approach*.

Other approaches to try

What else can be done?

- Change the function: additive function, polynomial function, etc.
- Change L_2 loss to say L_1-loss (or some other loss function).

$$L(Y, f(\mathbf{X})) = |Y - f(\mathbf{X})|.$$

- Minimizing $\mathbb{E}|Y - f(\mathbf{X})|$ we get the solution: $\widehat{f}(\mathbf{x}) = \text{median}(Y|\mathbf{X} = \mathbf{x})$.
- More robust, but computationally expensive.
- There are other variants as well, but L_2 remains popular.

Supervised learning: classification

- What happens for classification?
- Here we have qualitative, categorical **outputs** Y.
- Possibilities:
  - $Y \in \mathcal{Y} = \{0, 1\}$ or $Y \in \mathcal{Y} = \{-1, 1\}$ (Binary)
  - $Y \in \mathcal{Y} = \{1, \ldots, K\}$ (Multiclass)
  - We may have a multi-class problem dummy variable-coded as well.
- We may express loss functions as a $K \times K$ matrix where $K = |\mathcal{Y}|$.
  - $L(k, \ell)$ = price paid for (mis)classifying k as ℓ; if $k = \ell$, this is 0.
- Hence Risk (or EPE):

$$R(\widehat{Y}(\mathbf{X})) = \mathbb{E}[L(Y, \widehat{Y}(\mathbf{X}))] = \mathbb{E}_{\mathbf{X}} \sum_{k=1}^{K} [L(k, \widehat{Y}(\mathbf{X}))] \mathbb{P}(Y = k | \mathbf{X}).$$

- Pointwise minimization gives:

$$\widehat{Y}^*(\mathbf{x}) = \arg\min_{y \in \mathcal{Y}} \sum_{k=1}^{K} L(k, y) \mathbb{P}(Y = k | \mathbf{X} = \mathbf{x}).$$

Bayes classifier: the gold standard

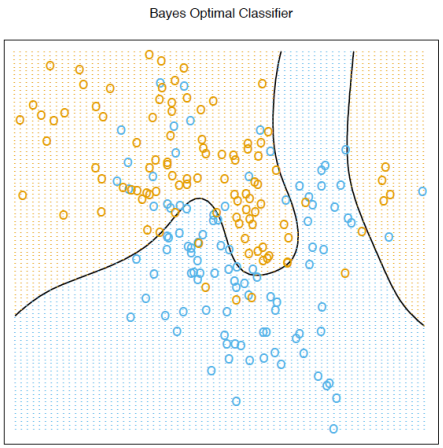
- Considering 0-1 loss function we have

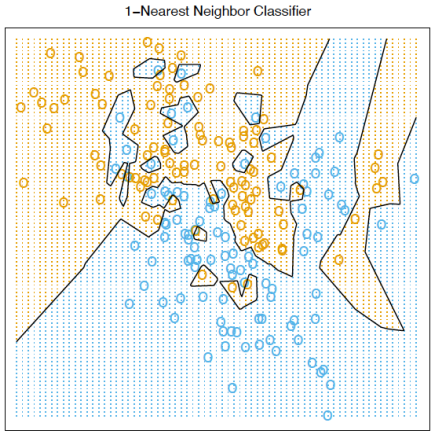
$$\widehat{Y}^*(\mathbf{x}) = \arg\min_{y \in \mathcal{Y}} [1 - \mathbb{P}(Y = y | \mathbf{X} = \mathbf{x})]$$

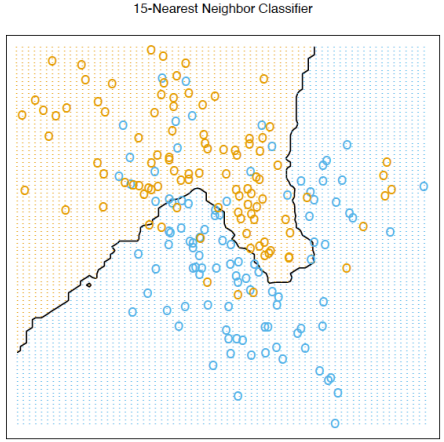
- Therefore $\widehat{Y}^*(\mathbf{x}) = k$ if $\mathbb{P}(Y = k | \mathbf{X} = \mathbf{x}) = \max_{y \in \mathcal{Y}} \mathbb{P}(Y = y | \mathbf{X} = \mathbf{x})$.
- This is quite intuitive.
- We call it the **Bayes classifier**.
- It classifies to the most probable class given the data.
- The error rate, i.e., EPE or Risk for the Bayes classifier $\widehat{Y}$ here is called the *Bayes risk* (also called *Bayes rate*).
- Recall the *Abalone gender classification* example.
  - What was the optimal classifier while minimizing error?

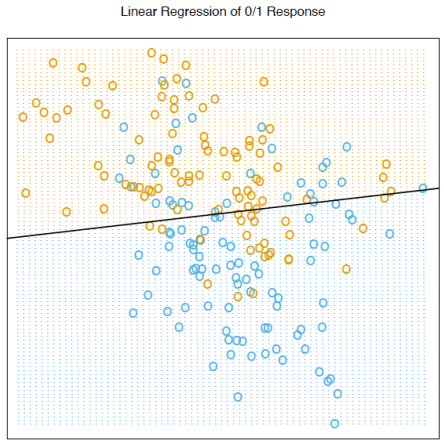
Bayes optimal decision boundary

- Naturally, in general we do not know $\mathbb{P}(\mathbf{X}, Y)$.
- k-NN classifier approximates this: replacing majority vote at a point to majority in its neighborhood.









- Left one is *Bayes optimal decision boundary* if we knew the generators.
- The next two are 1-NN and 15-NN classifiers from the data.
- Right one: binary variable with linearity assumption and squared error loss.
  - Also leads to the Bayes classifier theoretically.
  - Unfortunately, leads to counter-intuitive consequences, i.e, negative estimates of probability.

Empirical risk minimization (ERM)

- Issue 1: We do not have access to $\mathbb{P}(\mathbf{X}, Y)$.
- Issue 2: Searching over all functions f may be in-feasible.
- We have data:

$$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)$$

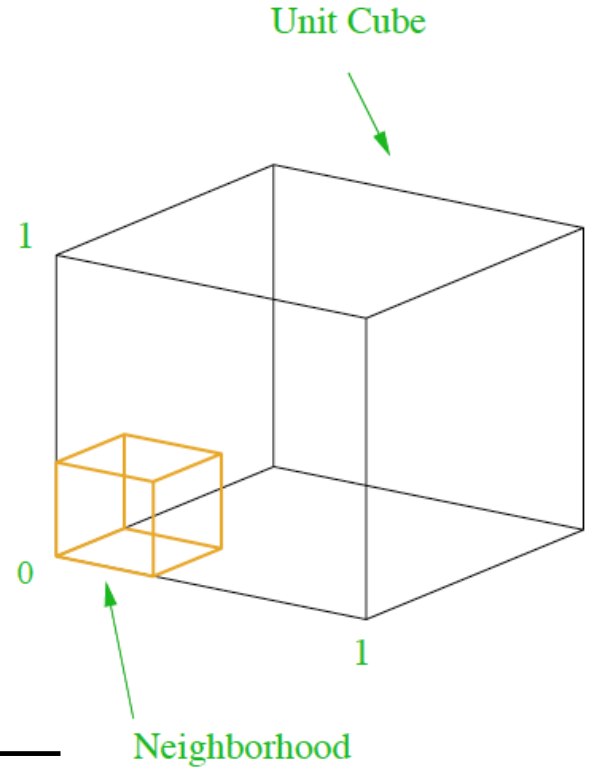
- So instead of minimizing risk $R(f)$ over all functions f we do the following
  - minimize empirical risk
  - search over a structured class of functions $f \in \mathcal{F}$ which is often parametrized, i.e., we may have $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$.

$$\min_{\theta \in \Theta} \widehat{R}(f_\theta) = \frac{1}{n} \sum_{i=1}^{n} L(y_i, f_\theta(\mathbf{x}_i)).$$

  - e.g., linear regression with squared-error loss (details in next class).

Curse of dimensionality

- Local methods do not perform well in high dimensions.
- Consider uniform distribution of points in the hypercube $[0,1]^p$ for $p \in \mathbb{N}$.
- Fix x_0 and say we want r proportion of points in its neighborhood.
- What is the length of such a neighborhood?
  - If Vol $= r$, edge length $= r^{1/p}$.



Volume	dimension	Edge length
0.01	10	0.63
0.01	20	0.79
0.1	10	0.79
0.1	20	0.89

- If 100 is a good sample size for inferring/estimating/learning with $p = 1$, a good sample size for $p = 10$ is 100^{10}.

Points to ponder

1. Under squared-error loss, what quantity should $f(\mathbf{x})$ estimate?
2. Under 0–1 loss, what determines the Bayes class?
3. If a nearest-neighbor method deteriorates after adding many irrelevant variables, what phenomenon is responsible?

References

- Hastie et al. (2001) *The Elements of Statistical Learning*.
- Hardt and Recht (2024) *Patterns, Predictions, and Actions*.